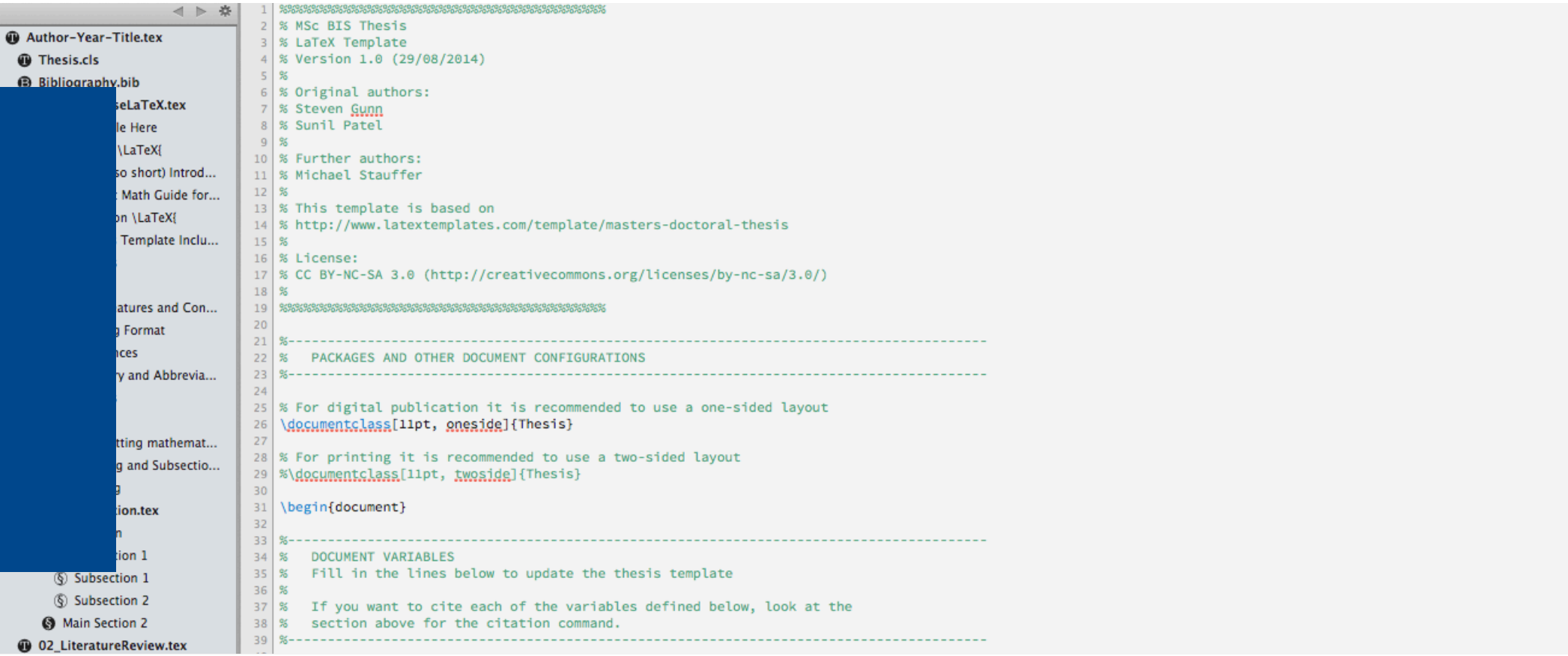
University of Applied Sciences and Arts Northwestern Switzerland
School of Business

Master Thesis Seminar

LaTeX Workshop



Content

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  - Download and Install SourceTree
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- LaTeX
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  - Download and Install Distribution
  - Download and Install Perl
  - Download and Install Editor
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- MSc BIS Template

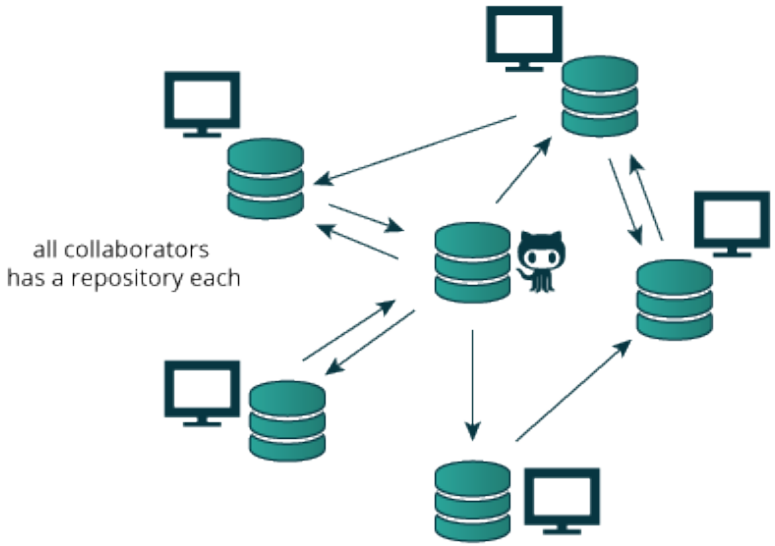
Git

Source: http://git-scm.com/images/logos/downloads/Git-Logo-2Color.png

Git – Basics (1/2)

What is Git?

- First release 2005
- Roots in the Linux Community
- Distributed Version Control System (DVCS)

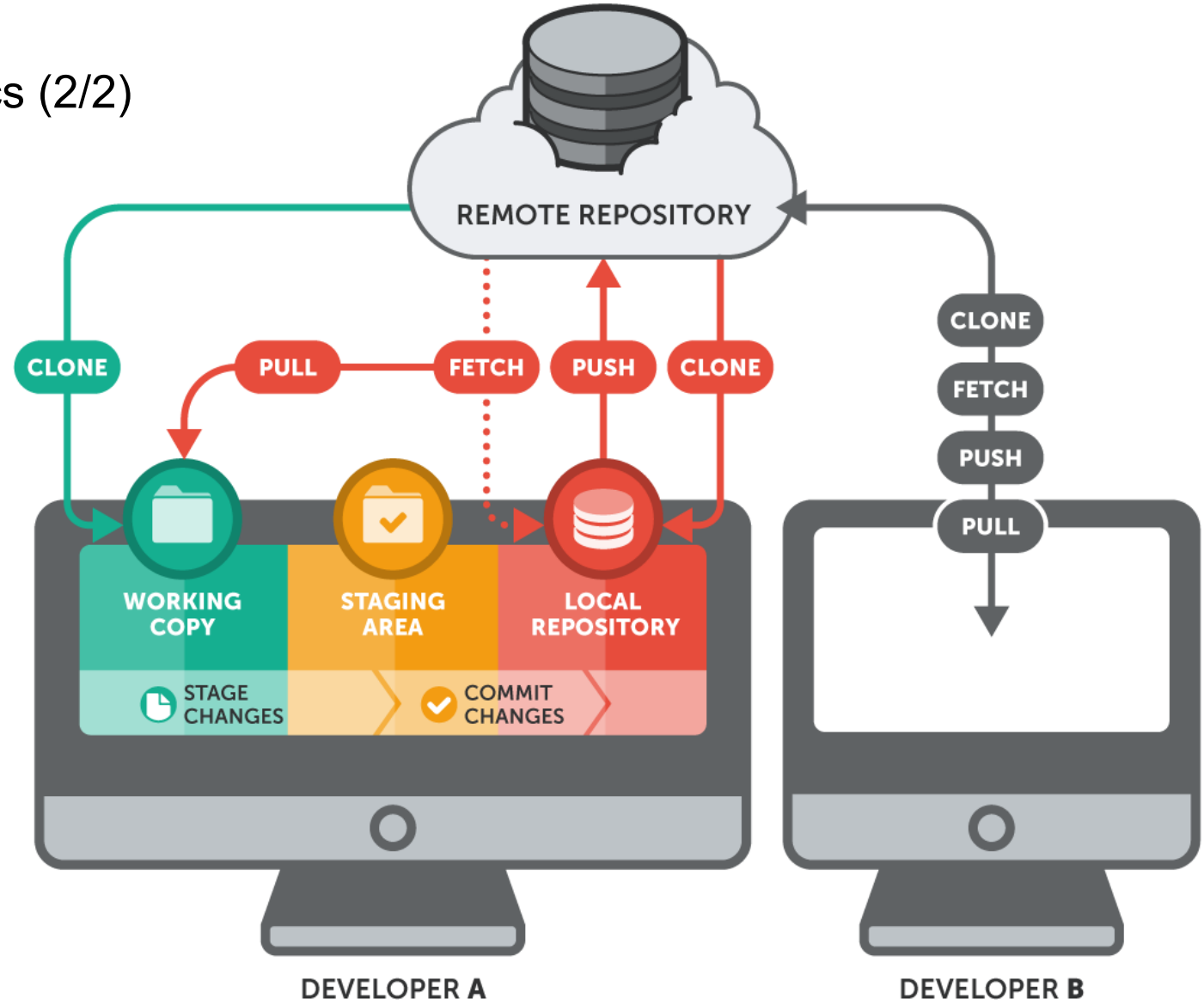


Why should I use Git?

- Allows collaboration
- Simplifies versioning (e.g. folder ThesisVersion_1.1)
- Acts as a backup

Source: http://invistruct.com/articles/explaining-basic-concepts-git-and-github/ http://git-scm.com/book/en/Getting-Started-A-Short-History-of-Git

Git – Basics (2/2)



Source: http://www.git-tower.com/learn/ebook/command-line/introduction#start

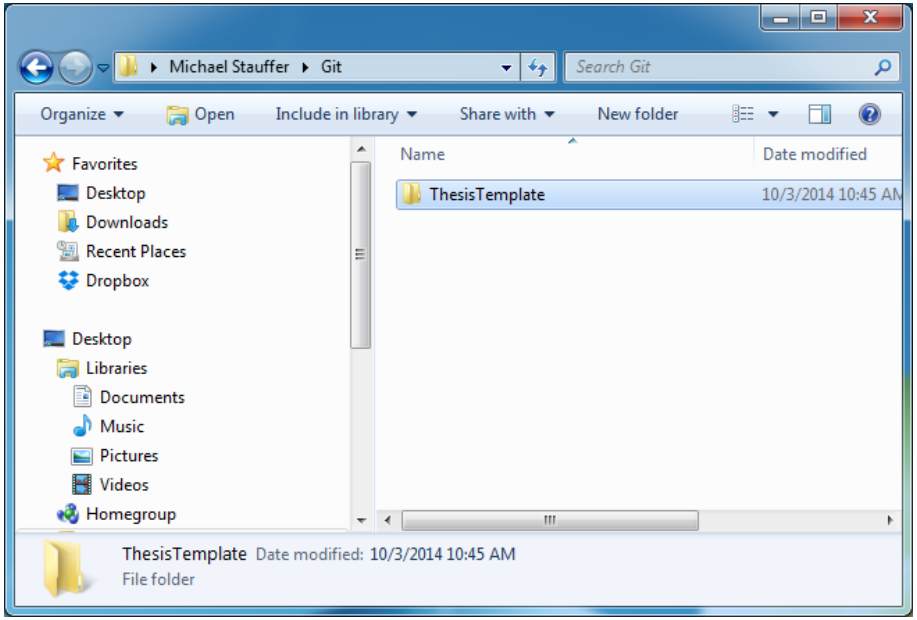
Git – Download and Install SourceTree

Is SourceTree running?

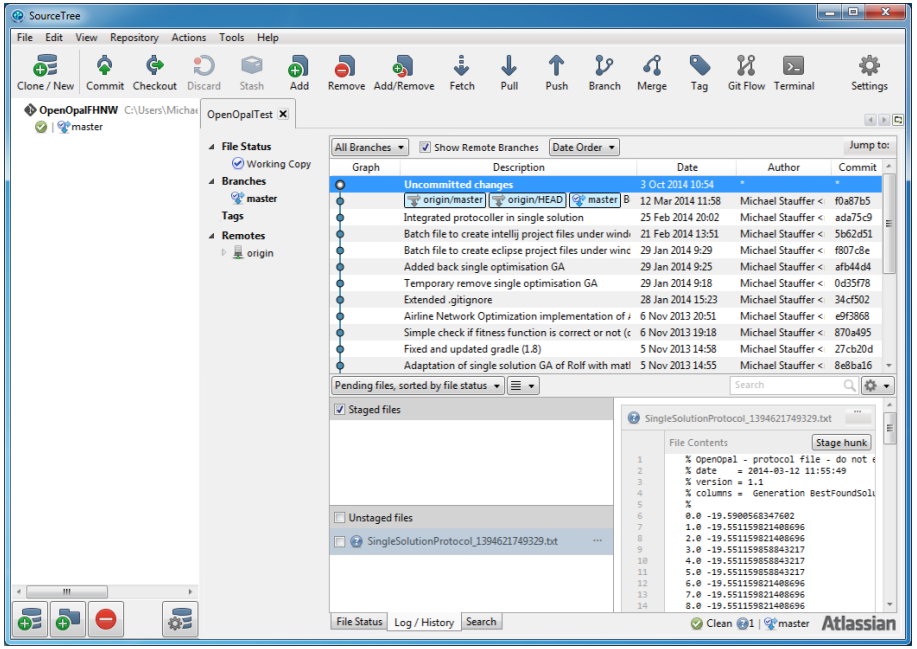
If not download it here http://www.sourcetreeapp.com/

Git – Access Repository (1/3)

1. Create local directory



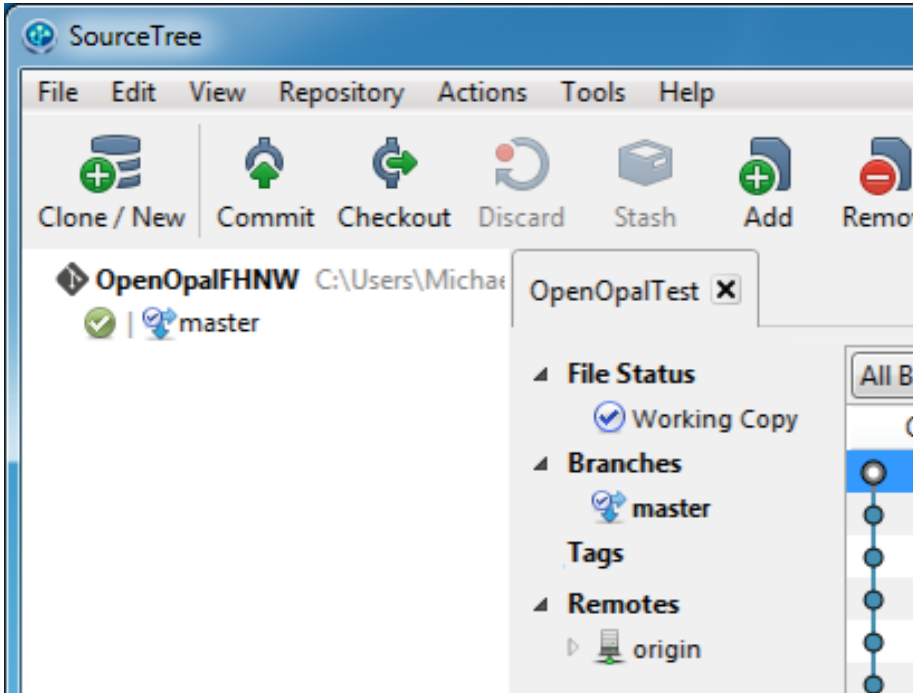
2. Launch SourceTree



Git – Access Repository (2/3)

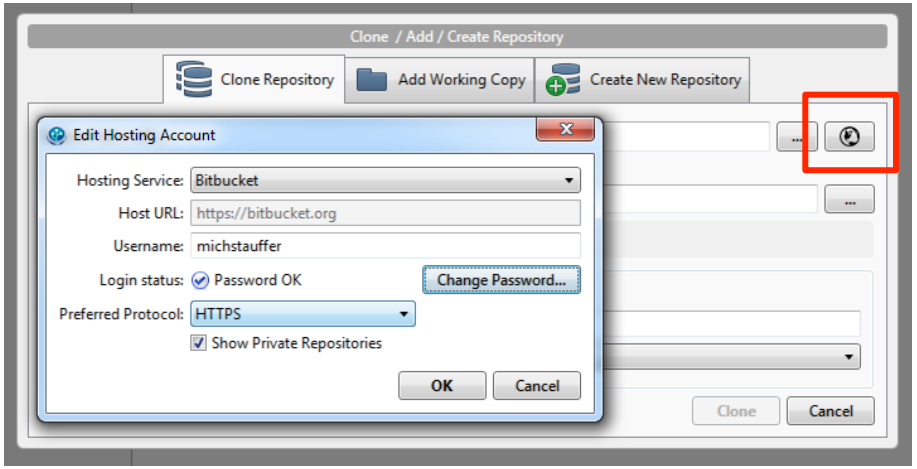
3. Add repository

Press “Clone / New”.



4. Add Bitbucket account

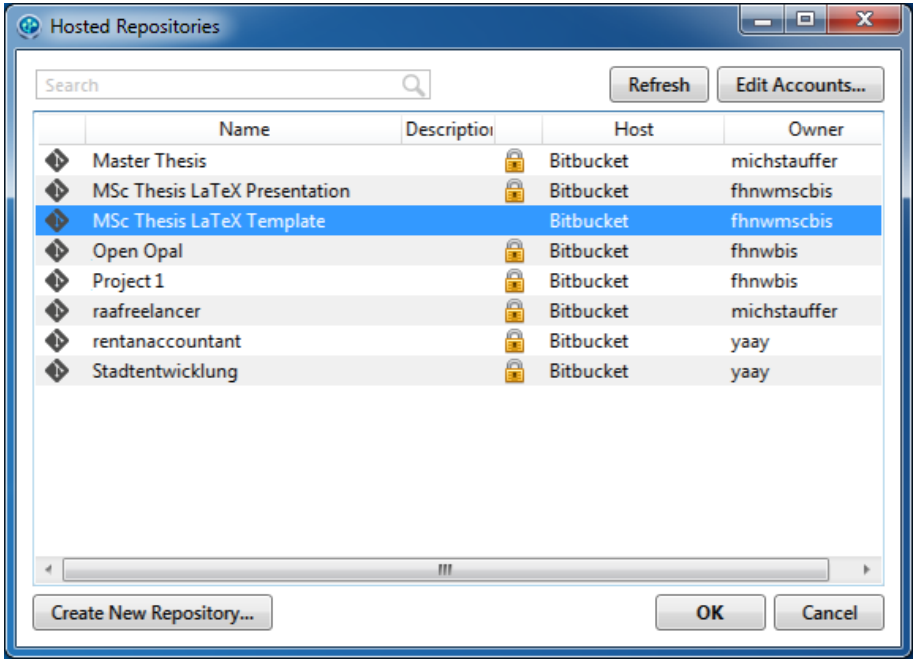
Press “Globe Icon” and enter your credentials. Press “OK”.



Git – Access Repository (2/3)

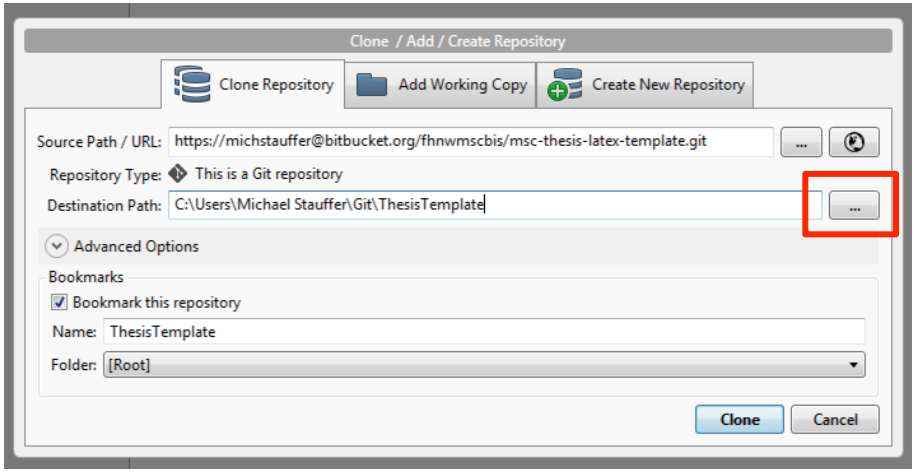
5. Select Project

Select “MSc Thesis LaTeX Template”. Press “OK”.



6. Clone Project

Change “Destination Path” to created directory. Press “Clone”



Git – Additional Resources

Git Cheat Sheet

http://www.clearvision-cm.com/wp-content/uploads/2014/08/Git-Cheat-Sheet_Clearvision.pdf

https://training.github.com/kit/downloads/github-git-cheat-sheet.pdf

Git Introduction

http://www.git-tower.com/learn/ebook/command-line/introduction#start

http://rogerdudler.github.io/git-guide/

Reference Documentation

http://git-scm.com/docs

Tutorial

https://try.github.io/levels/1/challenges/1 (highly recommended)

LaTeX

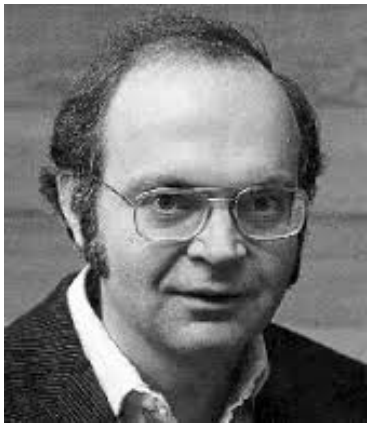
LaTeX

Source: http://commons.wikimedia.org/wiki/File:LaTeX_logo.svg

LaTeX – Basics (1/2)

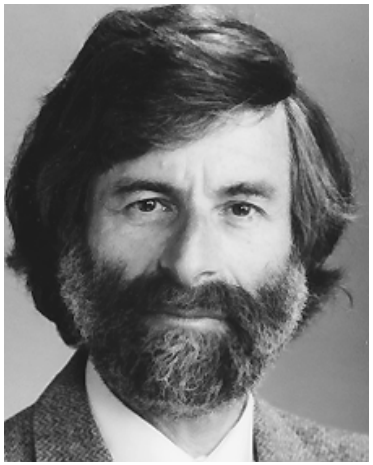
TeX (= typesetter)

- First release 1977 by Donald E. Knuth
- Typesetting engine for text and mathematical formulas
- TeX is pronounced “Tech”



LaTeX (= book designer)

- First release 1985 by Leslie Lamport
- Based on TeX
- Allows the use of predefined, professional layouts
- LaTeX is pronounced “Lay-tech” or “Lah-tech.”



Source: https://tobi.oetiker.ch/lshort/lshort.pdf, http://latex-project.org/intro.html

LaTeX – Basics (2/2)

WYSIWYG[1] approach (MS Word)

Author can directly see and modify the final look of a document while typing text.

Advantage

- Layout easy to change
- Suited for non techies

[1] **W**hat **Y**ou **S**ee **I**s **W**hat **Y**ou **G**et

LaTeX approach

Author cannot directly see and modify the final look of a document while typing text.

Advantage

- Professional layouts available
- Complex math. formulas
- Powerful citation
- Collaboration via Git
- Free

Source: https://tobi.oetiker.ch/lshort/lshort.pdf

LaTeX – Download and Install Distribution

Is LaTeX running?

If not download either http://miktex.org/ (MS) or http://www.tug.org/mactex/ (OS X)

LaTeX – Download and Install Perl

Is Perl running?

If not download here http://strawberryperl.com/ (Only necessary for MS user)

LaTeX – Download and Install Editor

Is TeXStudio running?

If not download it here http://texstudio.sourceforge.net/

LaTeX – Commands (Text)

Italic

```
Some of the greatest
discoveries in science were
made by \textit{accident}.
```

Some of the greatest discoveries in science were made by *accident*.

Bold

```
Some of the \textbf{greatest}
discoveries in science were
made by accident.
```

Some of the **greatest** discoveries in science were made by accident.

Underline

```
Some of the greatest
discoveries in
\underline{science} were made
by accident.
```

Some of the greatest discoveries in <u>science</u> were made by accident.

Source: https://www.sharelatex.com/learn/Bold,_italics_and_underlining

LaTeX – Commands (Headings)

Chapter

```
\chapter{Introduction}
```

Chapter 1
Introduction

Section

```
\section{Business Process
Management}
```

1.1. Business Process
Management

Subsection

```
\subsection{Design Time}
```

1.1.1 Design Time

LaTeX – Commands (Lists)

Unordered

```
\begin{itemize}
  \item First entry
  \item Second entry
\end{itemize}
```

- First entry
- Second entry

Ordered

```
\begin{enumerate}
  \item First entry
  \item Second entry
\end{enumerate}
```

1. First entry
2. Second entry

LaTeX – Commands (Figures)

Figure with caption and label

```
\begin{figure}[h]
    \centering
    \includegraphics[width=8cm]{mesh}
    \caption{A nice plot}
    \label{fig:mesh1}
\end{figure}
```

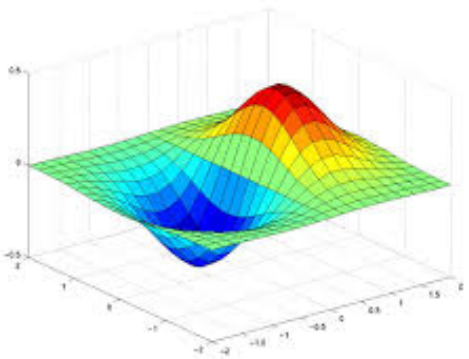


Figure 1: A nice plot

Referencing figure

```
In Figure \ref{fig:mesh1} you
can see a 3d plot.
```

In Figure 1 you can see a 3d plot.

Source: https://www.sharelatex.com/learn/Inserting_Images

LaTeX – Commands (Tables)

Table with caption and label

```
\begin{table}[h!]
\centering
\begin{tabular}{|c c c c|}
     \hline
     Col1 & Col2 & Col3 & Col4 \\
     \hline
     1 & 6   & 87837 & 787  \\
     2 & 7   & 78    & 5415 \\
     3 & 545 & 778   & 7507 \\
     4 & 545 & 18744 & 7560 \\
     5 & 88  & 788   & 6344 \\
     \hline

\end{tabular}
\caption{This is a table}
\label{table:1}
\end{table}
```

Col1	Col2	Co3	Col4
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table 1: This is a table

LaTeX – Commands (References)

Document

```
\usepackage{natbib}
\begin[colon]{document}

 LaTeX is great \cite{latexcompanion}.

\bibliography{Bibliography}
\end{document}
```

Bibliography.bib

```
@book{latexcompanion,

author    = "Michel Goossens",
title     = "The \LaTeX\ Companion",
year      = "1993",
publisher = "Addison-Wesley",
address   = "Reading, Massachusetts" }
```

```
LaTeX is great (Gossens, 1993).
```

References

M. Gossens. *The LaTeX Companion.* Addison-Wesley, Reading, Massachusetts, 1993.

Source: https://www.sharelatex.com/learn/Bibliography_management_with_natbib

It’s exercise time

Try to rebuild LatexExercise.pdf with the learned commands (see Moodle)

LaTeX – Additional Resources

LaTeX Cheat Sheet

http://www.stdout.org/~winston/latex/latexsheet.pdf

http://www.cis.upenn.edu/~byorgey/precalc/docs/LaTeX-cheat-sheet.pdf

LaTeX Introduction

https://tobi.oetiker.ch/lshort/lshort.pdf (highly recommended)

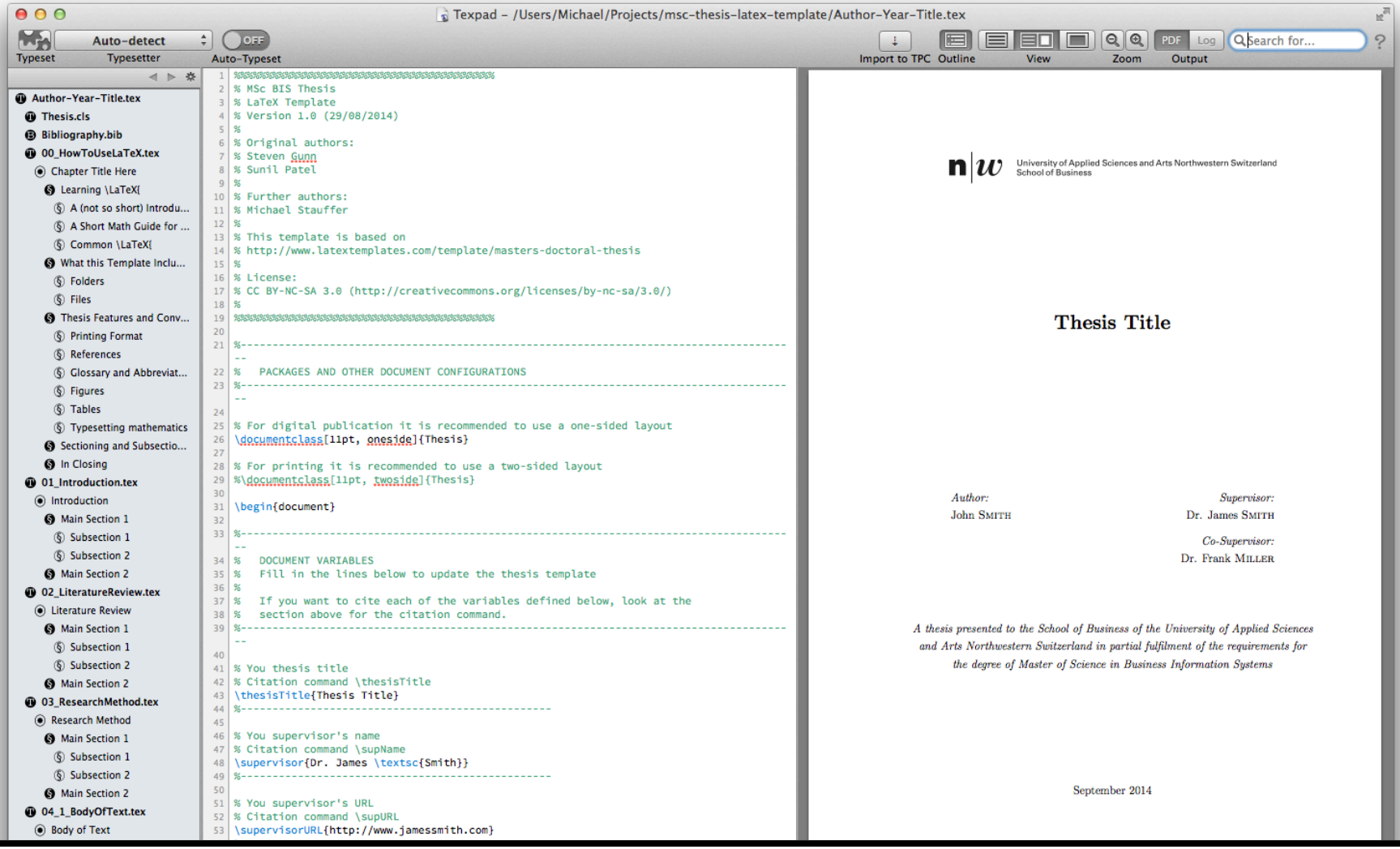
http://www.ctan.org/tex-archive/info/lshort/

Reference Documentation

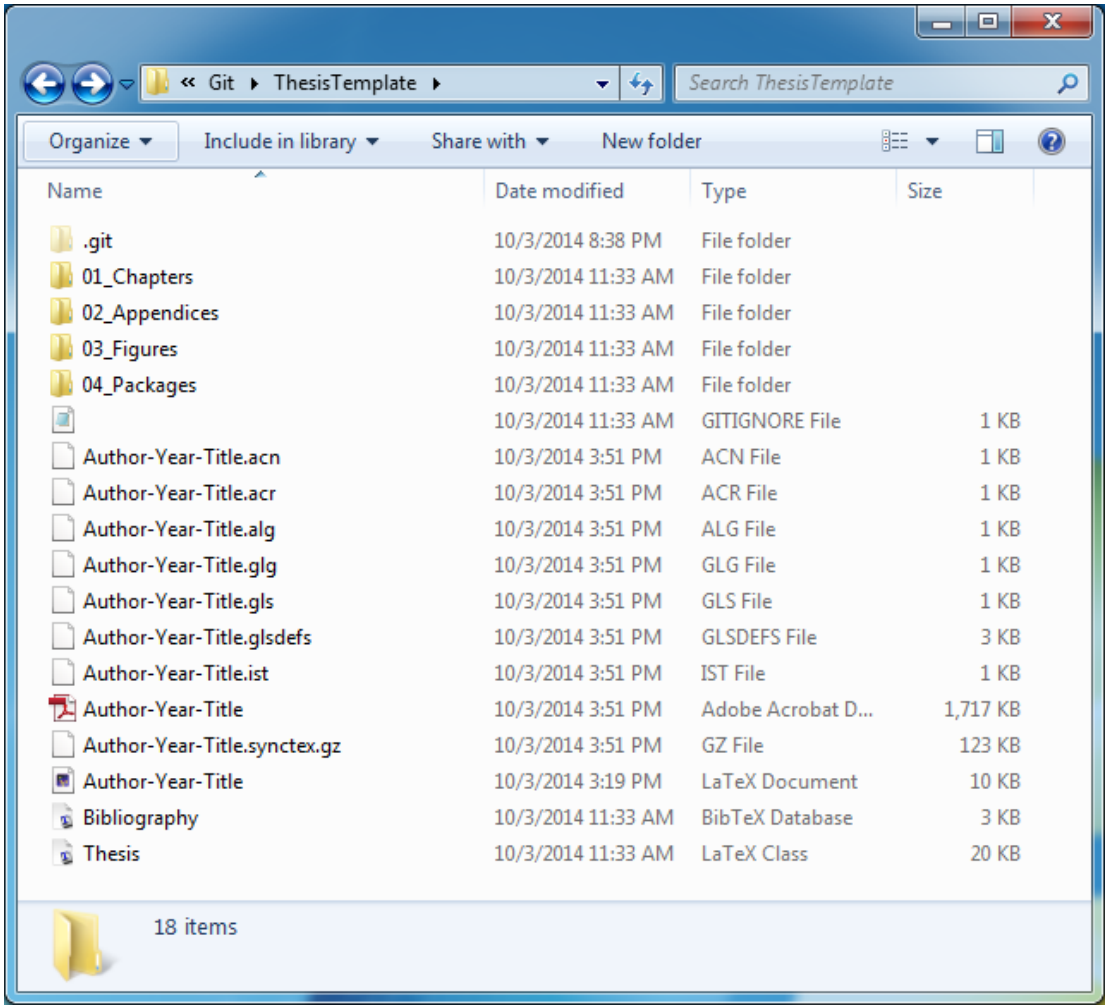
http://en.wikibooks.org/wiki/LaTeX

https://www.sharelatex.com/learn/Main_Page (highly recommended)

MSc BIS Template


Texpad – /Users/Michael/Projects/msc-thesis-latex-template/Author-Year-Title.tex
Typeset
Auto-detect
Typesetter
OFF
Auto-Typeset
Import to TPC
Outline
View
Zoom
PDF
Log
Output
Search for...
Author-Year-Title.tex
Thesis.cls
Bibliography.bib
00_HowToUseLaTeX.tex
Chapter Title Here
Learning \LaTeX{
A (not so short) Introdu...
A Short Math Guide for ...
Common \LaTeX{
What this Template Inclu...
Folders
Files
Thesis Features and Conv...
Printing Format
References
Glossary and Abbreviat...
Figures
Tables
Typesetting mathematics
Sectioning and Subsectio...
In Closing
01_Introduction.tex
Introduction
Main Section 1
Subsection 1
Subsection 2
Main Section 2
02_LiteratureReview.tex
Literature Review
Main Section 1
Subsection 1
Subsection 2
Main Section 2
03_ResearchMethod.tex
Research Method
Main Section 1
Subsection 1
Subsection 2
Main Section 2
04_1_BodyOfText.tex
Body of Text
%%%

MSc BIS Template – Thesis Structure



Open the directory you have created at the beginning

Keep in mind that **any figure** you would like to include needs to be in **`03_Figures`**

Open now Author-Year-Title.tex with TexStudio

MSc BIS Template – Compile Thesis

1. Change Configuration

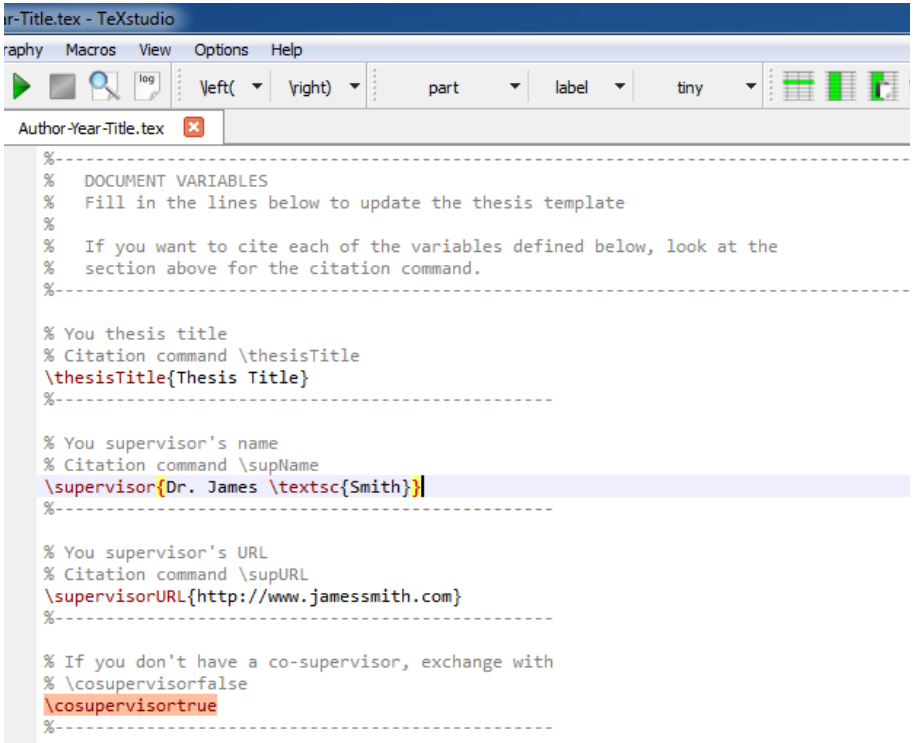
Go to “Options” > “Configure TeXStudio” > “Completion”

Enable the following .cwl-files

- `biblatex.cwl`
- `blindtex.cwl`
- `fanyhdr.cwl`
- `glossaries.cwl`
- `hyperref.cwl`
- `latex-document.cwl`
- `latex-mathsymbol.cwl`
- `tex.cwl`
- `thesis.cwl`
- `url.cwl`

2. Change Document Variables

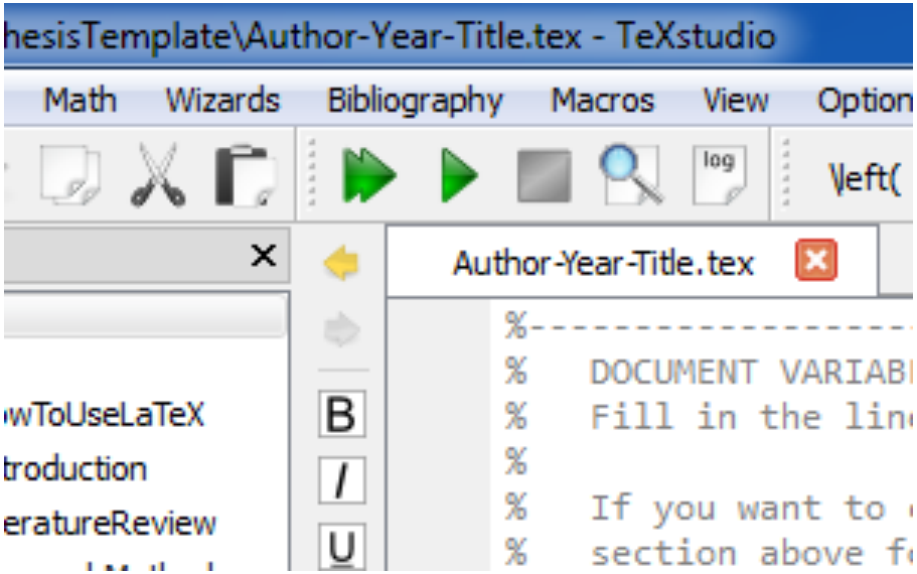
Save changes by “File” > “Save”



MSc BIS Template – Compile Thesis

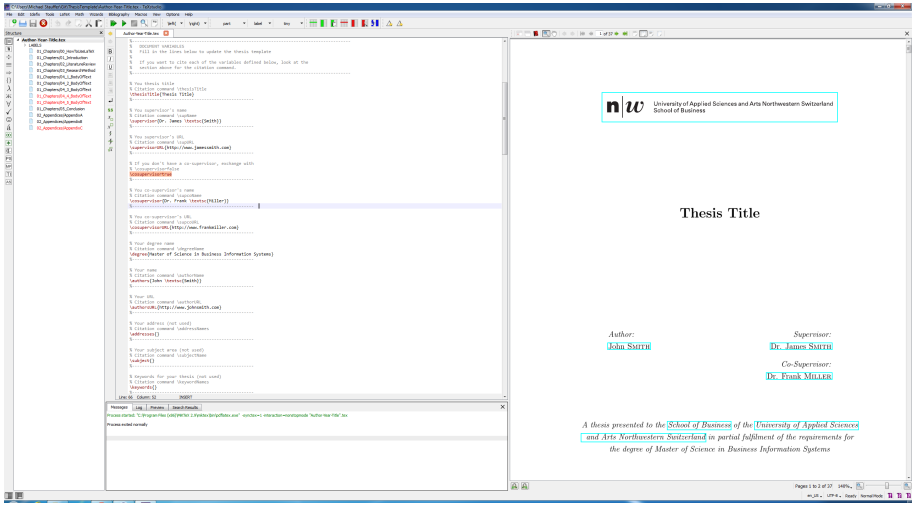
3. Compile Thesis

Press the green double arrow



4. Voilà

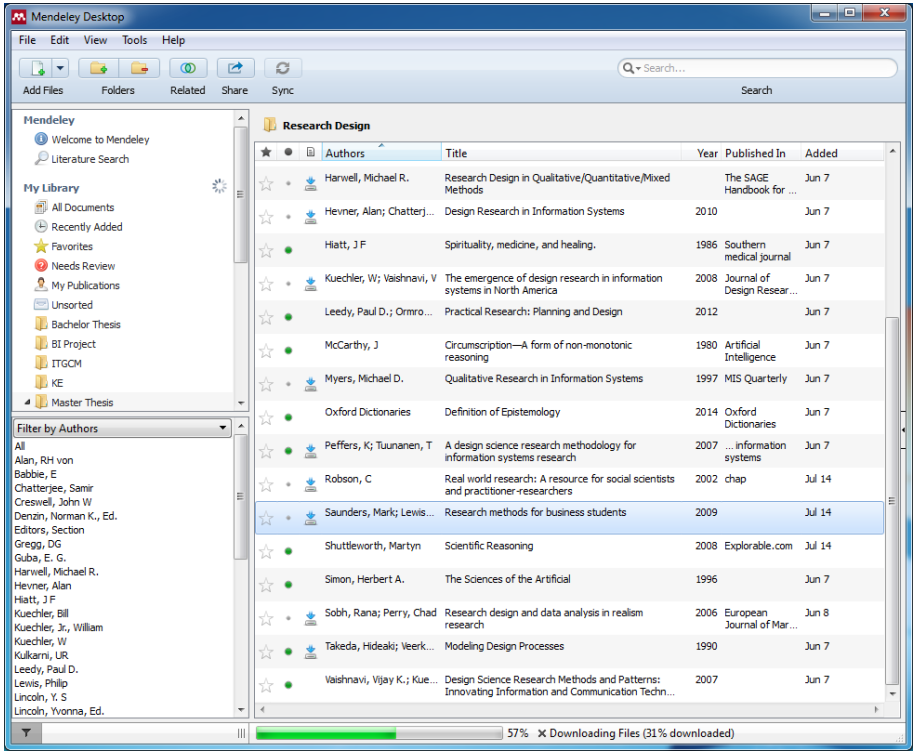
Author-Year-Title.pdf is created and ready to print



MSc BIS Template – Add references with Mendeley

1. Copy Citation

Select document in Mendeley and copy citation by “CTRL+K”



2. Paste Citation

Paste citation in LaTeX document by “CTRL+V”

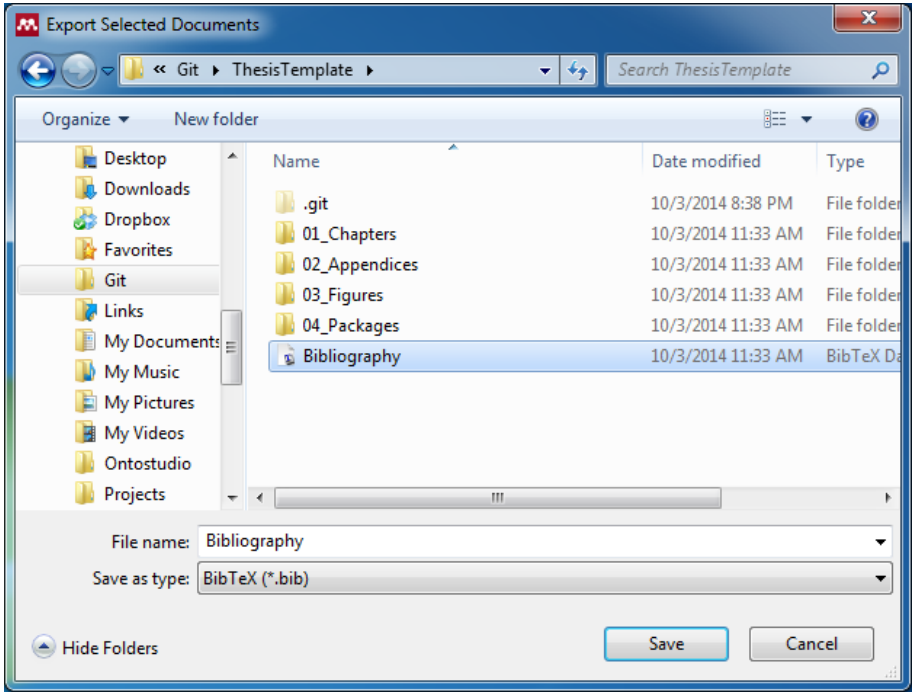
```
s in elit neque, quis varius elit. Phasellus fringilla,
lla auctor odio aliquet consequat. Ut nec nulla in ante
etium lorem. Pellentesque eget ornare odio. Proin
eque \cite{Saunders2009}.
```

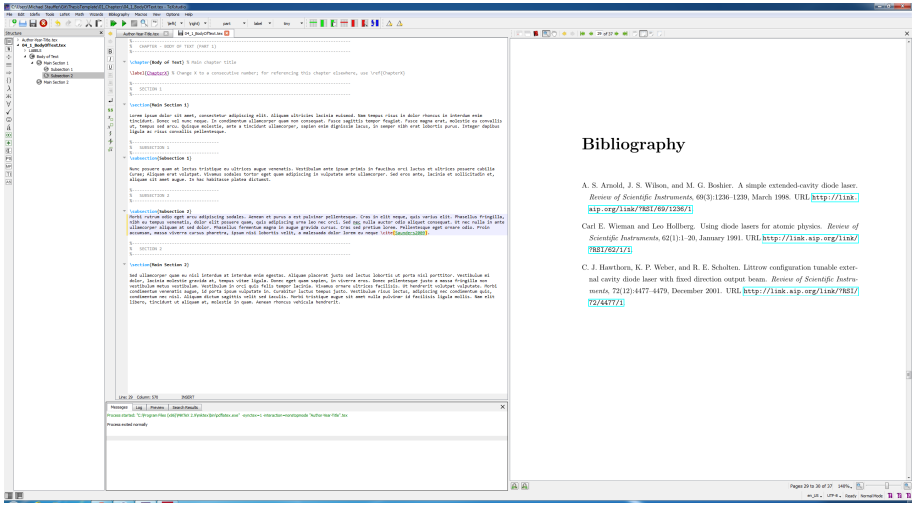
```
s lobortis ut porta nisl porttitor. Vestibulum mi
. Donec pellentesque justo a massa fringilla non
es facilisis. Ut hendrerit volutpat vulputate. Morbi
lum risus lectus, adipiscing nec condimentum quis,
nulla pulvinar id facilisis ligula mollis. Nam elit
```

MSc BIS Template – Add references with Mendeley

3. Export Citations

Select all cited documents, press “Export”, overwrite “Bibliography.bib”



4. Compile Thesis

Compile thesis again, new references should now be in bibliography.